

Total no. of Pages:02

Roll. No.:

1st SEMESTER, E. TECH (EE)

MID-SEMESTER EXAMINATION, February, 2025

Course Code: EEECC02

Course Title: Electronics Devices and Circuits

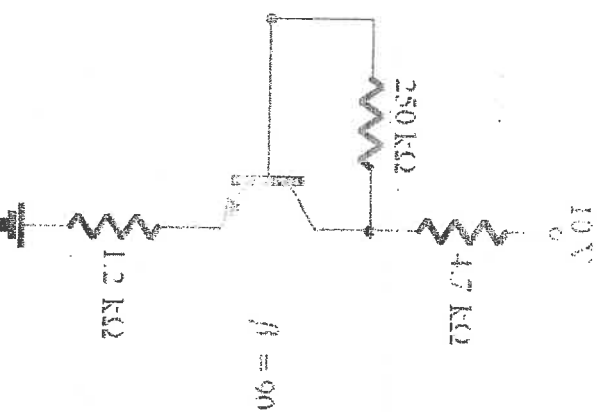
Time: 1:30 Hours

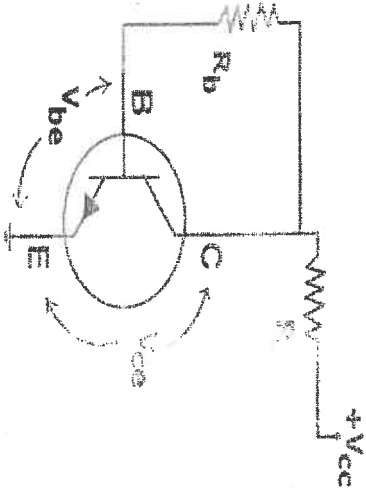
Note: Attempt all questions.

Max. Marks: 15

Q.no	Question	Marks	CO
1(a)	What is a p-n junction? Describe properties of an equilibrium p-n junction with the help of a diagram.	2	CO1
1(b)	The intrinsic carrier concentration of silicon sample at 300 K is $1.5 \times 10^{16}/m^3$. If after doping, the number of majority carriers is $5 \times 10^{20}/m^3$, what will be the minority carrier density?	1	CO1
2(a)	Draw a neat circuit diagram of bridge rectifier and explain its operation with output waveforms. Drive the average value of current and voltage.	2	CO1
2(b)	How can you measure the capacitance of Varactor Diode? State and explain with a suitable formula.	1	CO1
3(a)	Explain the various modes of operation of BJT with their corresponding applications.	2	CO2

3(ii)	An a.c. voltage of peak value 20 V is connected in series with a silicon diode and load resistance of 500Ω . If the forward resistance of diode is 10Ω , find (i) peak current through diode (ii) peak output voltage. What will be these values if the diode is assumed to be ideal?	2	CO2
4(i)	What is Base width modulation?	2	CO2
4(ii)	Determine the quiescent levels of I_C and V_{CE} for the given network. (Assume Silicon transistor)	1	CO2



5(a)	Draw and explain input, output, and transfer characteristics of CE-configuration.	2	CO2
5(b)	What is meant by bias stabilisation and stability factor? determine the expression of the stability factor for the circuit shown in figure below. 	1	CO2